

SciQuery: Architecture & Smart Caching Explainer

NCERT Class 10 Science Doubt-Solving Chatbot with Multi-Guard Semantic Cache

1. How the Chatbot Works

- ? Classification: User queries are classified into STANDALONE, FOLLOW_UP, or STYLE using zero-LLM heuristics.
- ? Follow-Up Rewrite: Ambiguous follow-ups are rewritten to standalone queries via LLM. Self-contained queries are kept unchanged.
- ? Retrieval: Retrieves top 6 passages from 734 textbook chunks via FAISS and bge-small-en-v1.5 embeddings.
- ? Out-of-Scope Detection: Queries scoring < 0.50 decline politely with empty citations, saving LLM calls.
- ? Generation: Answers generated strictly from NCERT context using Groq LLM with chapter citations.

2. How the Cache Works

- ? Latency Profile: Exact hit ~5 ms end-to-end (0.08 ms store), semantic hit ~11 ms end-to-end (5.8 ms store), fresh answer about 0.4 to 2 s normally and up to ~10 s when rate-limited.
- ? Exact Hash Match: Normalized queries perform an instant SQLite lookup.
- ? Semantic Match: Queries without numbers search a FAISS cache vector index (cosine threshold ≥ 0.88).
- ? Five Safety Guards: Number Guard (identical numbers/units), Symbol Guard (exact match of single letters like C, F, P and 2F), Contrast Guard (differentiates antonym pairs like concave/convex, acids/bases), Key-Term Guard (Jaccard ≥ 0.65 on content words), and Question-Type Guard (define vs law vs numerical).

3. What We Cache & Never Cache

- ? CACHED: Verified standalone NCERT answers with citations, canonical forms of follow-up questions, and contextual follow-up pairs.
- ? NEVER CACHED: Style requests (simpler, points), out-of-scope declines, numerical questions (semantic skipped), and error responses.

4. What Did Not Work

- ? Pure Cosine Similarity: Opposites score high (concave/convex = 0.9505; R=20 vs R=30 = 0.9244), causing hallucinations without guards.
- ? Strict Jaccard Threshold (0.80): Rejected natural phrasing variations; lowered to 0.65 to accommodate minor stopword differences.
- ? Global Style Detection: Heuristics without session history misclassified legitimate questions (e.g. combination reaction).

5. Architecture & Decision Flowchart

